

COMMODITY VIEWS

From Energy to Metals: Why to Still Diversify Into Commodities

- As energy flows start to normalize through the Strait of Hormuz, we take stock of how commodities have performed year to date, highlighting that the energy supply shock triggered by the Iran conflict is only the most recent of several examples seen over the past year of why strategic investment portfolios can benefit from diversifying into commodities.
- Investing in different commodities can help diversify risks inherent to equity/bond portfolios under different circumstances, including (1) commodity supply shocks, like the Hormuz disruption, which might lead to higher inflation and lower economic growth, (2) structural demand support for commodities that face challenges to growing supply, and (3) a flight to real assets when fiscal sustainability or other financial risks arise. We see many of the drivers behind such circumstances, which are supportive of commodity returns, remaining relevant going forward.
- In particular, we think that the Iran conflict ultimately reinforces many of the themes supporting power and metals demand, more so than oil and gas. From a potential increased reliance on EVs, to further investment into renewable power generation, both of which require further grid investment, to potentially larger defense spending, and growing competition to win the AI race, these themes are highly supportive of power, copper, lithium and aluminum demand.
- Combined with the fact that power infrastructure can face bottlenecks and that metals refining remains highly geographically concentrated, this strong structural support to power and metals demand leaves these markets vulnerable to tightening shocks. As a result, while oil and gas have traditionally had a bigger impact on inflation, and its supply concentration and associated geopolitical risks remain, sharp price increases might start to also appear more often across power and industrial and precious metals markets going forward.

Samantha Dart
+1(212)357-9428 |
samantha.dart@gs.com
Goldman Sachs & Co. LLC

Daan Struyven
+1(212)357-4172 |
daan.struyven@gs.com
Goldman Sachs & Co. LLC

Lina Thomas
+1(212)902-8376 | lina.thomas@gs.com
Goldman Sachs & Co. LLC

Yulia Zhestkova Grigsby
+1(646)446-3905 |
yulia.grigsby@gs.com
Goldman Sachs & Co. LLC

Hongcen Wei
+1(212)934-4691 |
hongcen.wei@gs.com
Goldman Sachs & Co. LLC

Lavinia Forcellese
+44(20)7774-9243 |
lavinia.forcellese@gs.com
Goldman Sachs International

Aurelia Waltham
+44(20)7051-2547 |
aurelia.waltham@gs.com
Goldman Sachs International

Filippo Cuscito
+44(20)7051-9073 |
filippo.cuscito@gs.com
Goldman Sachs International

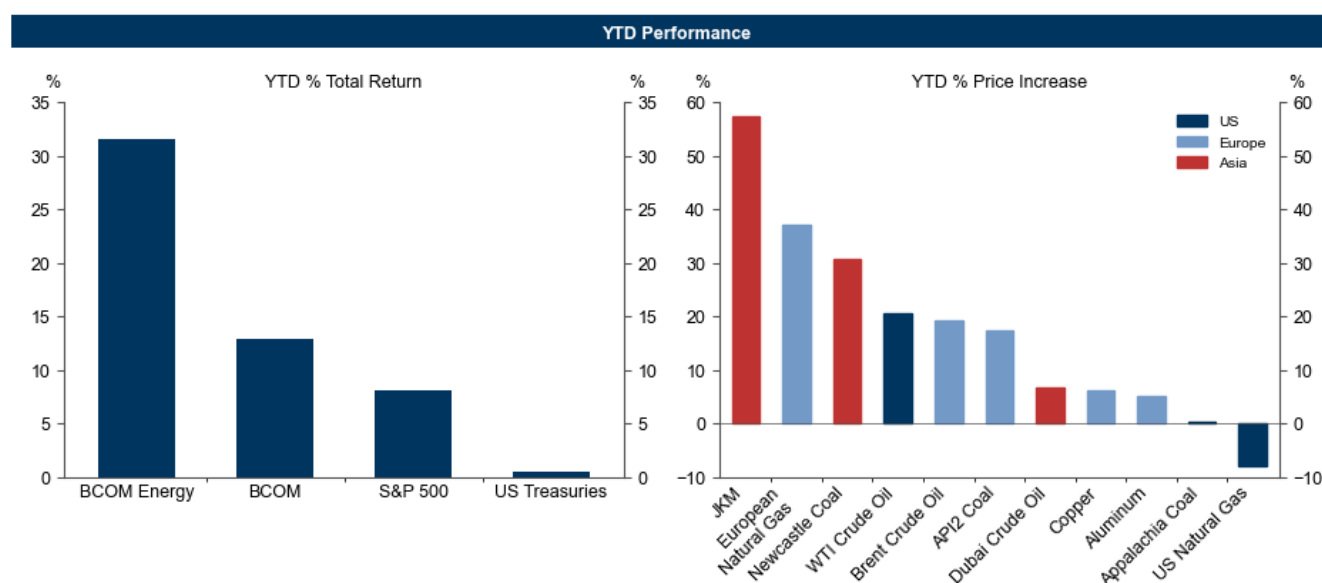
Laura Cyr
+1(212)902-3435 | laura.x.cyr@gs.com
Goldman Sachs & Co. LLC

Alexandra Paulus
+1(212)902-7111 |
alexandra.paulus@gs.com
Goldman Sachs & Co. LLC

Commodities Can Help Diversify Equities/Bonds' Risks Under a Broad Range of Circumstances

1. A large energy shock, the second in only five years. While the 2022 energy crisis impacted mostly natural gas supply, lifting liquefied natural gas (LNG) prices across Europe and Asia, this year's 16-week Strait of Hormuz disruption significantly reduced supply of both oil and gas, causing a broad increase in energy prices, while threatening the global economic outlook. Even as crude oil prices now recede following the US-Iran deal to re-open the Strait of Hormuz, broad commodity price gains year to date, combined with a strong oil roll yield¹ during the conflict, have supported year-to-date commodity returns well above those of equity and bonds (Exhibit 1), though with higher volatility.

Exhibit 1: The Strait of Hormuz disruption contributed to commodity returns outperforming other assets year to date, especially as a strong roll yield for oil combined with higher prices



Source: Bloomberg, Goldman Sachs Global Investment Research

Importantly, the upside to energy prices during the conflict was limited by what proved to be a more flexible global market than what we originally expected. In particular, the sharp drop in China liquefied natural gas (LNG) imports in Mar/Apr, the first two months of the conflict, and in oil imports to this day, helped limit global market tightness. Still, by May, three months into the conflict, crude oil and oil product (like gasoline, diesel and jet fuel) prices had rallied 43% and 63%, respectively, vs pre-war levels, while European gas and Asia LNG prices rallied 50% and 70%. And while crude oil prices have come off sharply over the past couple of weeks, the same cannot be said of gasoline or diesel, where we expect demand to rebound faster than supply, keeping balances tight for longer, and prices slower to come down.

Overall, assuming that the re-opening process continues, the Hormuz shock will likely have a limited scarring effect on global economic activity and hence on energy demand. Our economists expect global GDP growth at 2.4% this year, 0.4pp down from 2025.

¹ The roll yield is the gain or loss generated by the investor "rolling" their position from an expiring futures contract into a new contract with a later expiration date. The roll yield is positive when the expiring contract is more expensive than the futures contract they are rolling into (i.e., when the forward curve is in backwardation).

Had the Iran conflict lasted longer, its negative impact on global growth might have been significantly larger at -2pp from our pre-war expectations, underscoring the potential value of diversifying into commodities in strategic investment portfolios.

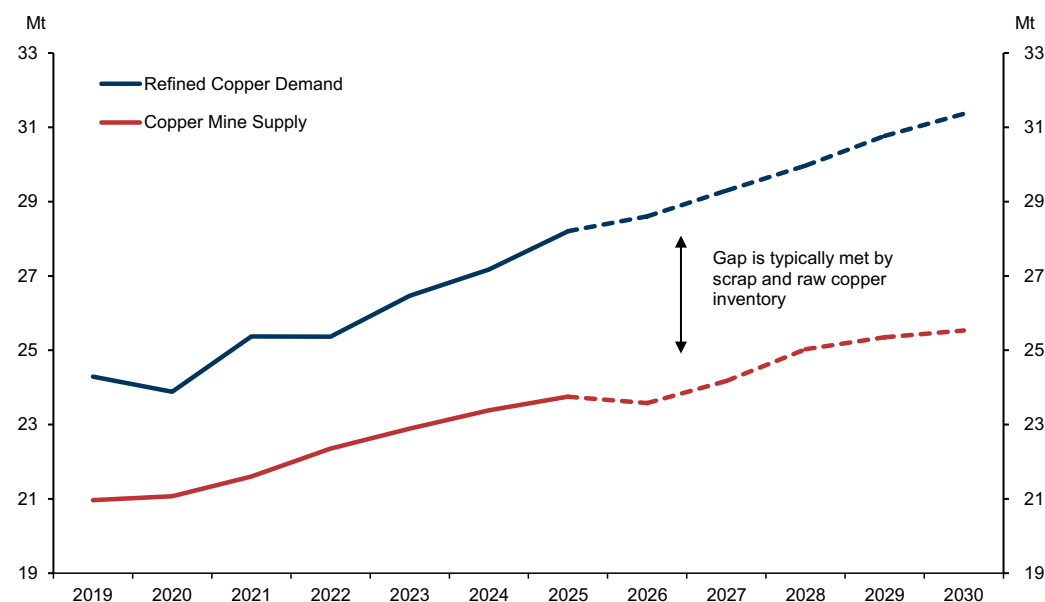
2. Long-term structural support for copper prices, with a near-term policy boost.

The pairing of strong structural support to copper demand with a long-cycle supply side, where it can take an extended period of time for supply to respond to price incentives, sets the stage for a well supported price environment for copper.

On the demand side, growth has been increasingly tied to strategic sectors, with grid and power infrastructure likely driving over 60% of copper demand growth by 2030 (vs. 2025), alongside direct boosts from defense, electric vehicles, renewable generation capacity, and data centers. Importantly, we see investment in aging Western power grids as a national security priority, due to its critical role in AI and energy security. This shifts copper demand from being mainly cyclical to increasingly strategic, making it less sensitive to economic slowdowns and high prices than traditional demand sources such as construction or white goods.

At the same time, copper supply might struggle to grow fast enough ([Exhibit 2](#)). Multiple recent [mine incidents](#) highlight the growing structural challenges in copper mining as copper mines get deeper, grades get lower and ore gets harder, requiring greater investment. This adds to operating costs and sustaining capex, and limits the ability of supply to respond quickly. As a result, copper prices have to rise to solve this gap by incentivizing higher scrap supply and potential demand substitution into alternative materials such as aluminum.

Exhibit 2: We expect copper demand growth to continue to outpace that of mined supply

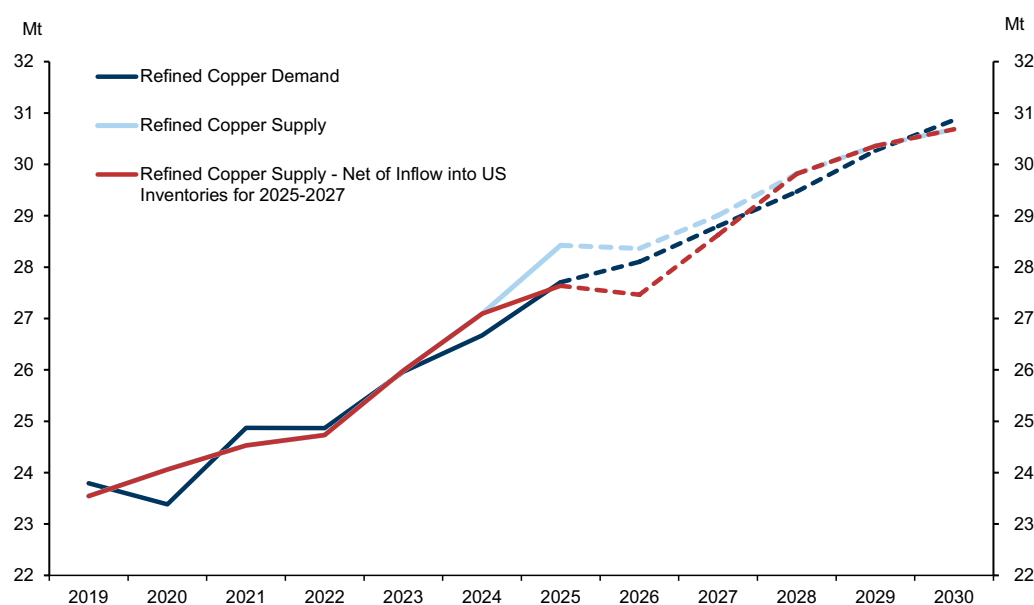


Source: Wood Mackenzie, CRU, S&P Global Market Intelligence, SMM, ICSG, Goldman Sachs Global Investment Research

While the copper balance tightening we expect is gradual in nature, shifts in policy have brought some of that tightness forward. Specifically, since the US imposed import tariffs on steel and aluminum last year, copper imports into the US picked up significantly in anticipation of a potential US copper import tariff being implemented. While there remains no US tariffs on refined² copper imports, copper tonnes have continued to head to the US, causing the ex-US copper balance to tighten significantly ([Exhibit 3](#)), and prices to briefly reach an all-time high of over \$14,000/t in May, before moderating to the low-to-mid \$13,000s.

These tighter copper balances led us to recently raise our end-2026/average 2027 LME copper forecasts to \$13,735/\$13,800. Our longer-term view remains that prices will need to rise further by the next decade to balance the market. By 2035, we think a price of \$15,000/t will be needed to keep aging mines in operation, lift scrap collection rates, deliver additional substitution and support the development of new mines.

Exhibit 3: Policy-risk-driven copper flows into the US drove the ex-US market into a deficit this year



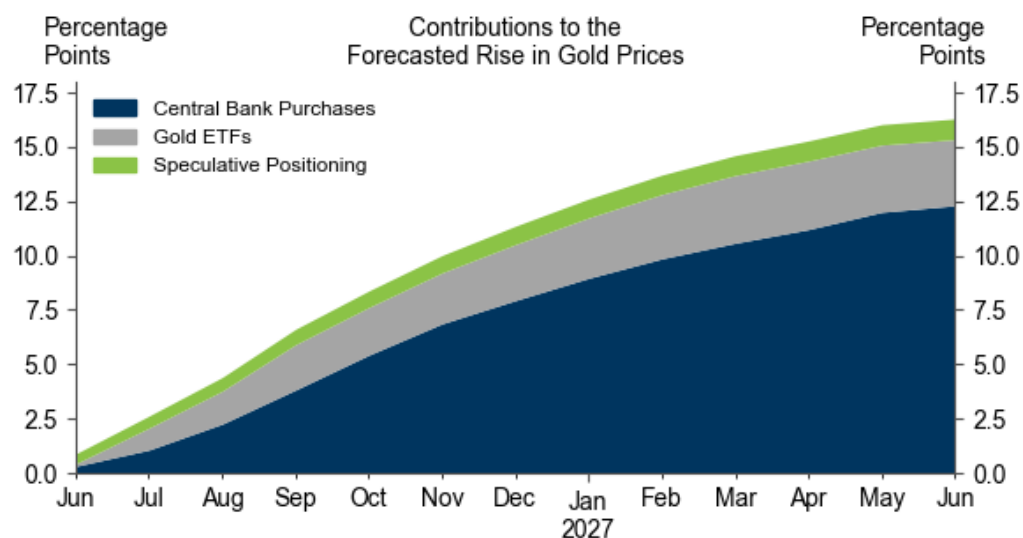
Source: Wood Mackenzie, CRU, SMM, ICSG, S&P Global Market Intelligence, Goldman Sachs Global Investment Research

3. Gold is not done. Despite gold's strong rally since 2022 (up 123%), we continue to see further upside, driven by both structural and eventually cyclical factors. Structurally, EM central bank diversification—following the 2022 freezing of Russia's reserves—remains the anchor of our \$4,900/toz end 2026 forecast ([Exhibit 4](#)). While the pace of central bank gold purchases has moderated to ~50 tonnes/month on a 3-month (seasonally adjusted) and 12-month moving average basis, we view the ongoing diversification trend as structural. A recent World Gold Council survey supports our view: a record 45% of the 76 central banks surveyed between February and May expect to increase their own gold reserves over the next 12 months, while ~90% expect global reserves to rise with the remainder expecting broadly stable holdings ([Exhibit 5](#)). We therefore assume continued central bank accumulation of 50t/month in 2026 and 40t/month in 2027.

² The US implemented a 50% Section 232 tariff on semi-finished copper products and copper-intensive derivative products effective August 1, 2025. Refined copper, ores, concentrates and scrap were excluded.

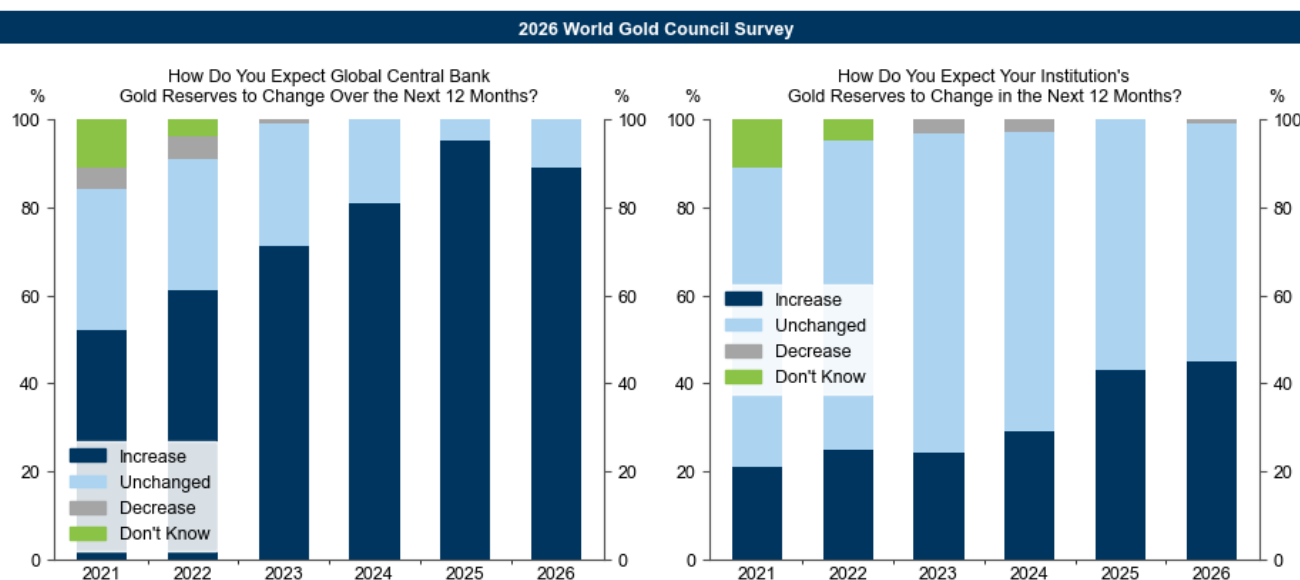
Cyclically, however, gold faces near term headwinds from the potential hit to demand for macro policy hedges (as a hawkish Fed helps fade the debasement theme) and as markets are pricing in Fed hikes this year amid inflation concerns, weighing on rate-sensitive ETF demand. We expect these headwinds to at least partly reverse over time. While we assume stable macro-policy hedges demand for gold, we expect ETF positioning to gradually rise, consistent with our economists' view of a delayed but ongoing easing cycle in 2027H2. Over the medium term, risks to our gold price forecast remain skewed to the upside on net. Gold's share in private portfolios remains small, and the Iran episode — together with broader geopolitical developments (e.g., Greenland, Venezuela) — may eventually accelerate private diversification into gold, including by weighing on perceptions of Western fiscal sustainability.

Exhibit 4: Continued central bank diversification remains the main driver of our constructive gold price outlook



Source: Goldman Sachs Global Investment Research

Exhibit 5: The recent World Gold Council central bank survey suggests continued gold demand



Source: World Gold Council, Goldman Sachs Global Investment Research

To Each (Inflation Shock) Their Own (Inflation Hedge)

While the Hormuz shock has illustrated how commodities exposure can work as a hedge against inflation, it also demonstrated with the sharp sell off in gold in the period that not all commodities work equally well as a hedge under any inflation shock. In practice, inflation typically arises through three distinct regimes – late cycle inflation, supply disruptions, and institutional credibility risk – and each calls for a different hedge.

Regime #1: Late Cycle – Hedge With Cyclical Commodities

When the business cycle runs hot, equities initially benefit from strong growth. But as the economy begins to outpace its productive capacity – what economists call a positive output gap – inflation pressures build and real bond returns weaken. Over time, rising input costs compress margins and equity growth starts to soften. It is at this point – when bond prices weaken and equity returns begin to lose momentum – that commodities tend to provide diversification through stronger returns. Commodity performance typically strengthens late in the cycle because a positive output gap implies demand exceeds supply. In commodity markets, this imbalance shows up as sustained inventory drawdowns. Late in the cycle, inventories have been drawing for long enough to approach depletion, pushing prices higher – especially for cyclical commodities like oil and industrial metals.

Regime #2: Supply Disruption – Hedge With a Broad Commodity Basket (Ex. Precious Metals)

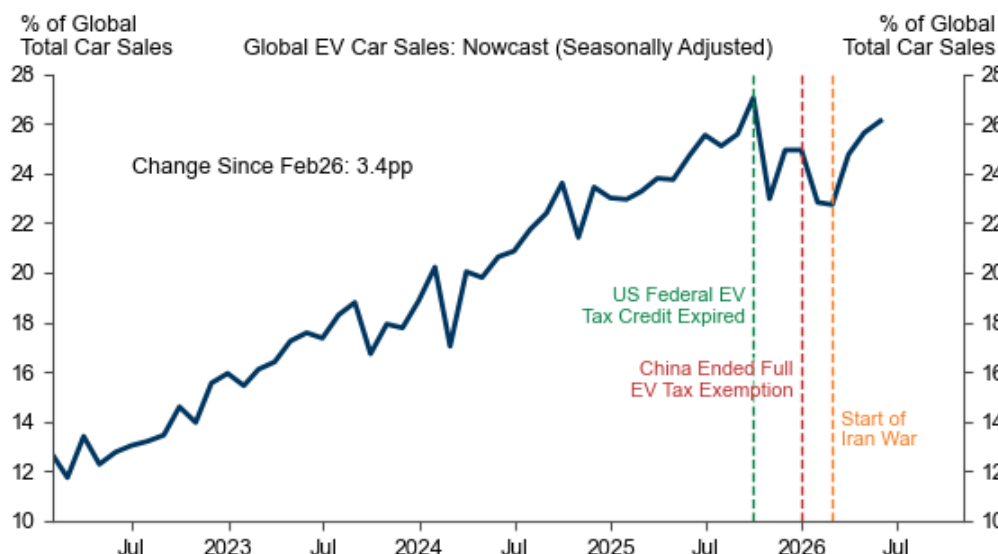
When a supply disruption hits – as just exemplified during the Iran conflict – inflation rises while growth weakens, weighing on bond and equity prices simultaneously. Commodities, as the disrupted input, are then among the few assets to deliver positive real returns. Because the source and timing of disruptions are inherently unpredictable, a broad commodity basket (ex. precious metals) offers the most robust protection.

Regime #3: Institutional Credibility Risk – Hedge With Gold

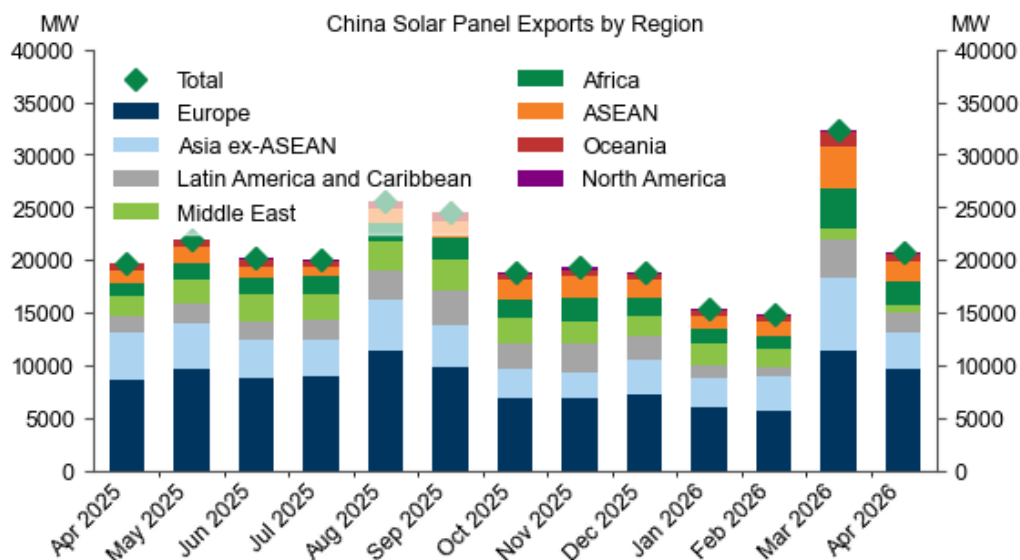
In the first two inflation regimes – late cycle inflation and supply disruptions – gold is not an effective hedge. If anything, gold often sells off initially: higher inflation can lead markets to price in rate hikes, raising the opportunity cost of holding a non yielding asset, while equity drawdowns can trigger margin call liquidations in gold, whose liquidity makes it a ready source of cash. Gold hedges a narrow inflation regime: when inflation expectations rise due to concerns around institutional credibility or macro policy, causing bonds and equities to sell off together in real terms. Gold then stands apart as the key neutral asset whose value does not rely on any government backing.

The Iran Conflict Appears on Its Way to a Resolution. What Next?

While oil demand will rebound in the aftermath of the Iran conflict, including what we expect will be an over 1 mb/d push for the rebuilding of strategic petroleum reserves, we think that the Iran conflict ultimately reinforces many of the themes supporting power and metals demand in particular. From a potential increased reliance on EVs, to further investment into renewable power generation, both of which require further grid investment, to potentially larger defense spending, and growing competition to win the AI race, these themes are highly supportive of power, copper, lithium and aluminum demand, while potentially weighing on long-term demand growth of oil and natural gas (Exhibit 6 and Exhibit 7).

Exhibit 6: Global EV sales have sharply accelerated during the Iran conflict...

Source: Government Data, IEA, Goldman Sachs Global Investment Research

Exhibit 7: ...as have China solar panel exports

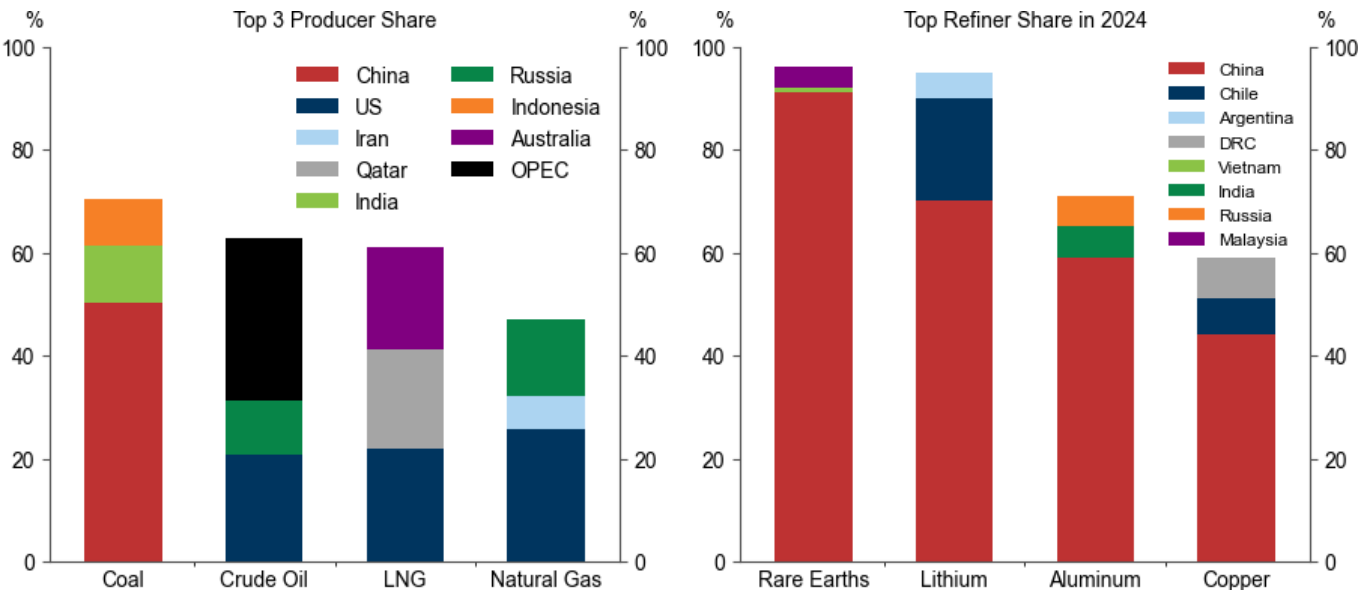
Source: Ember, Goldman Sachs Global Investment Research

Combined with the fact that power infrastructure can face bottlenecks, as has been the case in the US Mid-Atlantic power market, and that metals refining remains highly geographically concentrated (Exhibit 8), this strong structural support to power and metals demand leaves these markets vulnerable to tightening shocks.

As a result, while oil and gas have traditionally had a bigger impact on inflation, and their supply concentration and associated geopolitical risks remain, sharp price increases might start to also appear more often across power and industrial and precious metals markets going forward. In particular, our economists have flagged that higher industrial metals prices and China's export controls on rare earths and related products were among the main drivers of the recent acceleration in the US' core producer price index,

while expecting that higher US electricity prices will continue to boost consumer electricity inflation.

Exhibit 8: Commodity supply is currently heavily concentrated, with a large share of supply in geopolitical or trade-dispute hot spots



Source: EIA, IEA, WoodMac, CRU, Goldman Sachs Global Investment Research

Appendix

Exhibit 9: The recent Hormuz energy supply disruption is the latest example of why it pays to diversify into commodities

Total returns

	Dollar Weight	Historical Performance				GS Forecast	
		2023	2024	2025	YTD	Balance of Year	2027
BCOM	100	-8	5	16	15	3	1
BCOM ex. Agriculture & Livestock	63	-10	8	22	19	7	1
S&P GSCI	100	-4	9	7	26	7	1
S&P GSCI ex. Agriculture & Livestock	76	-3	11	11	37	10	0
Energy	52	-5	10	-5	55	10	-2
Industrial Metals	14	-4	3	29	7	4	-2
Precious Metals	10	12	26	69	-8	17	12
Agriculture	14	-8	0	-8	0	-3	2
Livestock	9	0	20	27	5	3	1

We highlight the commodity indices ex Agriculture & Livestock because we currently don't forecast agriculture and livestock prices, and instead use forwards.

Source: Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

We, Samantha Dart, Daan Struyven, Lina Thomas, Yulia Zhestkova Grigsby, Hongcen Wei, Lavinia Forcellese, Aurelia Waltham, Filippo Cuscito, Laura Cyr and Alexandra Paulus, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Samantha Dart Goldman Sachs & Co. LLC, Daan Struyven Goldman Sachs & Co. LLC, Lina Thomas Goldman Sachs & Co. LLC, Yulia Zhestkova Grigsby Goldman Sachs & Co. LLC, Hongcen Wei Goldman Sachs & Co. LLC, Lavinia Forcellese Goldman Sachs International, Aurelia Waltham Goldman Sachs International, Filippo Cuscito Goldman Sachs International, Laura Cyr Goldman Sachs & Co. LLC, Alexandra Paulus Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: <https://www.goldmansachs.com/worldwide/india/research-analyst>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors

should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.

EUROPEAN ECONOMICS ANALYST

UK—The Economic Cost of Brexit, Ten Years On

- The UK economy has materially underperformed since the 2016 EU referendum, shifting from a growth profile closer to the US to one closer to the Euro area. Our updated “Doppelgänger” analysis suggests real GDP is around 6% below its counterfactual path, more than our previous 5% estimate two years ago. While these figures could capture the impact of other factors as well as Brexit, robustness checks point to an underperformance of at least 4%. These estimates are broadly consistent with other studies. Model-based estimates of the cost of leaving the EU average around 5%, while recent empirical research finds a 6–8% impact.
- Analysis of the channels through which Brexit could have affected output also points to a significant drag. Goods trade volumes have underperformed by 10–15% since the referendum. Services trade has held up better, but recent studies nonetheless find a meaningful Brexit-related hit. Lower trade intensity has likely reduced productivity, lowering GDP by 2–4%. Business investment has also been weak since 2016—with business surveys suggesting this is partly due to Brexit—likely lowering output by 2%. The impact on labour supply through migration has been more mixed, with lower EU immigration but higher non-EU inflows. That said, shifting migration patterns have probably contributed to labour shortages in some sectors.
- Taken together, our estimates indicate that Brexit has materially weighed on Britain’s economic performance relative to other advanced economies. While new trade agreements could, in principle, help mitigate some of the costs of Brexit, the benefit from new (non-EU) trade deals is likely to be small.
- That said, we see two reasons why the peak Brexit headwind might now be behind us. First, estimates point to a front-loaded drag, suggesting that the Brexit growth headwind should ease over time. Second, opinion polls now show greater support for a closer EU relationship, and the government is pursuing more EU-UK cooperation. An ambitious reset—such as closer single-market or customs-union alignment—could reverse a meaningful share of the Brexit cost and support stronger trend growth. Consistent with this, we expect average growth in the UK to pick up from 1.2% since 2016 to 1.5% over the coming years.

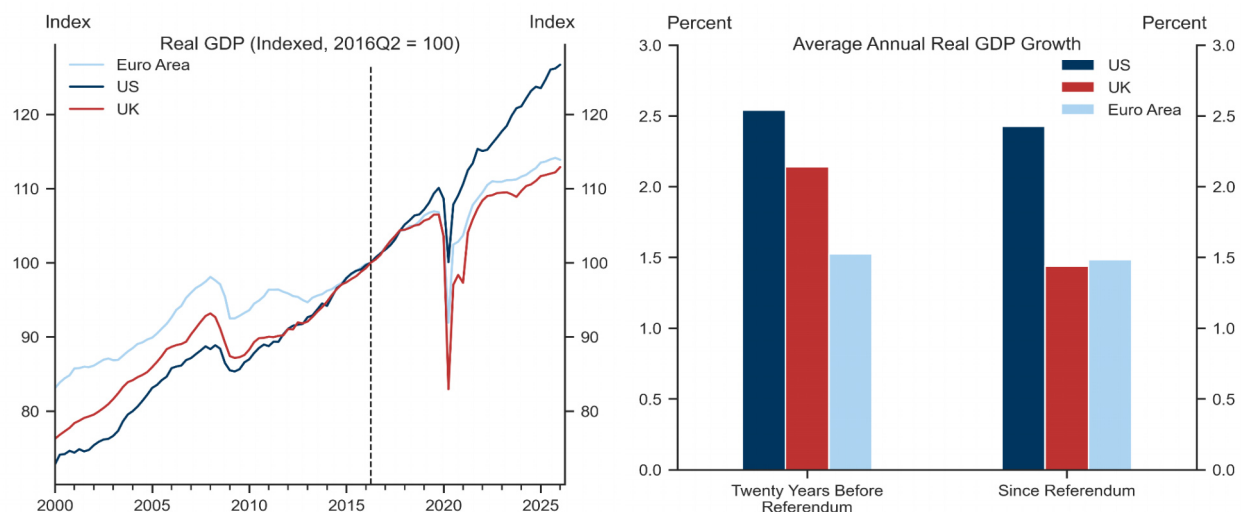
James Moberly
+44(20)7774-9444 |
james.r.moberly@gs.com
Goldman Sachs International

Sven Jari Stehn
+44(20)7774-8061 | jari.stehn@gs.com
Goldman Sachs International

UK—The Economic Cost of Brexit, Ten Years On

The UK voted to leave the EU ten years ago this week and has underperformed other advanced economies significantly since then (Exhibit 1). UK growth looked more similar to the US in the twenty years leading up to the referendum, but more similar to the Euro area in the subsequent ten years.

Exhibit 1: UK Growth Was Closer to the US Before Brexit, More Like the Euro Area Since Then



Source: Goldman Sachs Global Investment Research, Haver Analytics

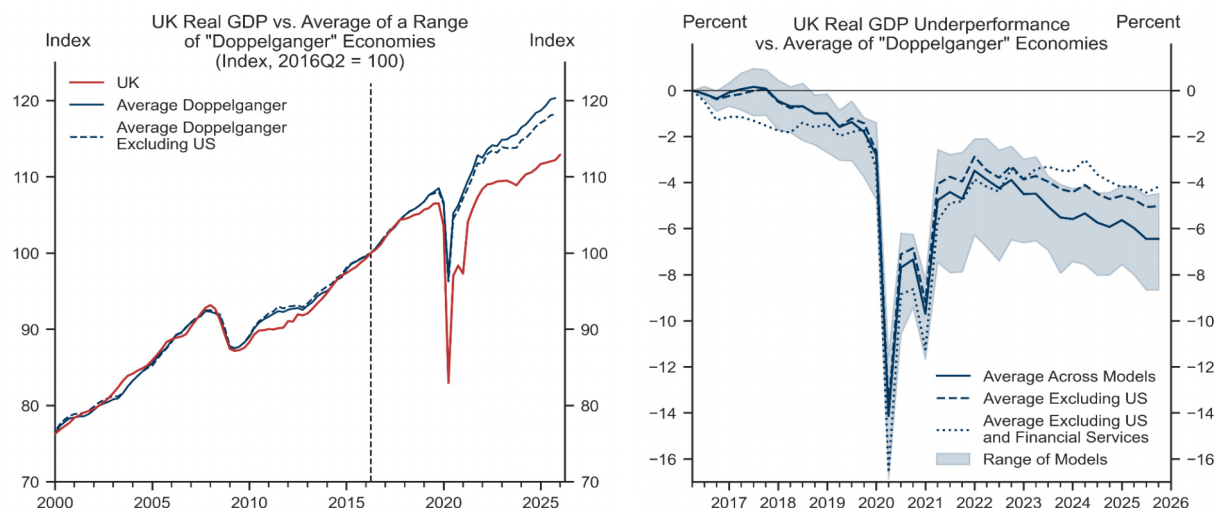
Given the ten-year anniversary of the referendum, we take stock of the macroeconomic costs of the Brexit vote and update our 2024 estimates.

Counting the Cost

We start by examining empirically how much UK real GDP has underperformed other advanced economies since 2016 using a “Doppelgänger” analysis.¹ This method uses statistical techniques to find the best combination of other countries that match the path of UK real GDP before the referendum and uses this same combination to project what might have happened thereafter.

Our estimates suggest that the UK has fallen short of its Doppelgänger by 6% since the referendum, compared with the 5% we estimated two years ago (Exhibit 2, left). Varying the group of countries in the analysis—including the G7, G10, OECD, OECD Europe—points to a range of 4–8% for the output cost (Exhibit 2, right).

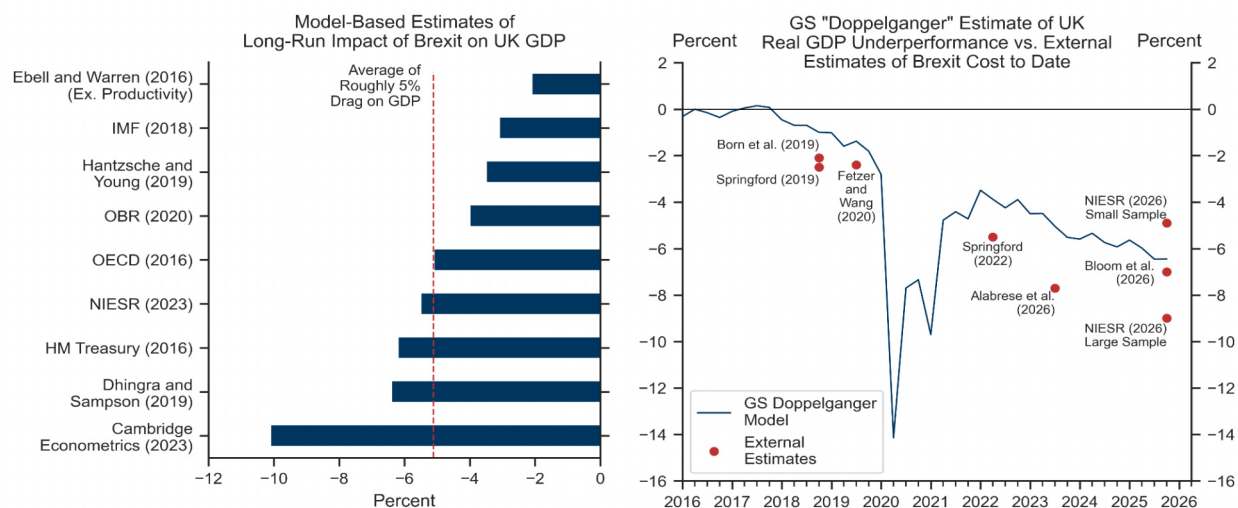
¹ This synthetic control method was applied to Brexit by Benjamin Born and co-authors in “The Costs of Economic Nationalism: Evidence from the Brexit Experiment,” May 2019.

Exhibit 2: Our Doppelgänger Model Points to Significant Underperformance Since the Referendum

Source: Goldman Sachs Global Investment Research, Haver Analytics

A shortcoming of the Doppelgänger approach is that not all of the post-2016 underperformance can be attributed to Brexit, as other structural changes have affected Britain's relative economic performance. In particular, the inclusion of the US could skew the estimates, as the US has benefited from a much larger technology boom than other advanced economies during this time. Excluding the US from the Doppelgänger analysis points to a smaller output cost at 5% (dashed line in Exhibit 2, right). It is also possible that UK growth was temporarily boosted by a booming financial sector before the referendum relative to other advanced economies. Re-running the analysis on UK GDP without the financial sector and again excluding the US points to 4% underperformance, which is likely to be a lower-bound estimate (dotted line in Exhibit 2, right).

Our Doppelgänger estimates are consistent with prior studies for the economic cost of Brexit. Model-based studies—which simulate the economic repercussions of greater trade frictions outside of the EU—point to a total output cost of Brexit of around 5% (Exhibit 3, left). The latest empirical studies—often using Doppelgänger models—point to a hit of 6-8%, which has grown notably in recent years (Exhibit 3, right). We provide a more detailed summary of these studies in the appendix.

Exhibit 3: Studies Point to a Large (and Rising) Cost of Brexit

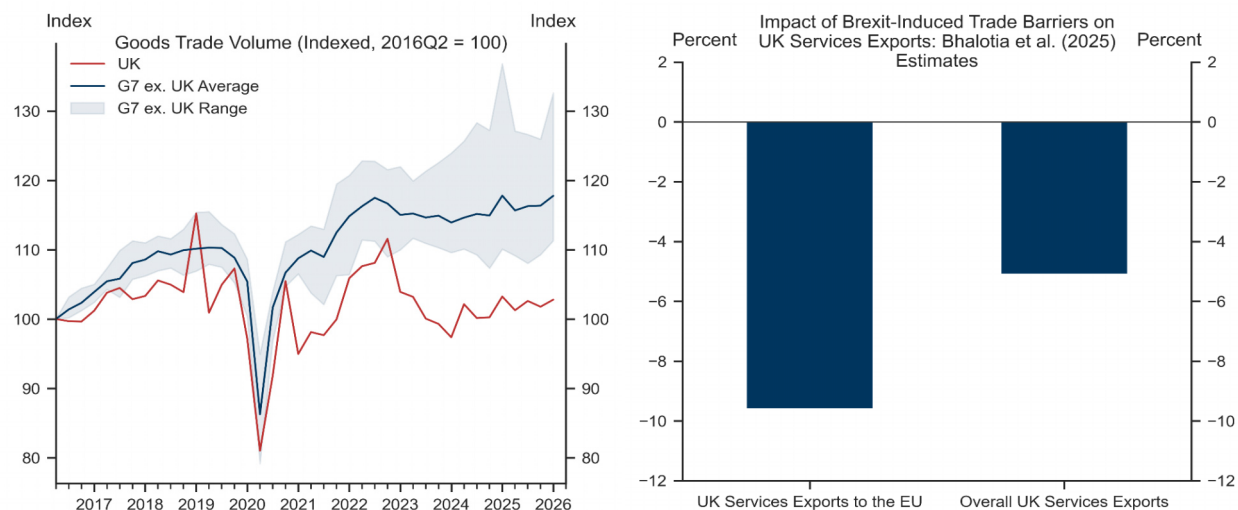
Ebell and Warren (2016) note that incorporating productivity losses would substantially increase their estimates.

Source: Goldman Sachs Global Investment Research, Various academic studies

Assessing the Channels

Given uncertainty around the top-down estimates, we consider the mechanisms of Brexit to help gauge how much of the UK's underperformance can be attributed to Brexit. Studies generally consider three mechanisms through which Brexit could have lowered long-term output: (1) international trade, (2) business investment and (3) immigration.

First, the UK has materially underperformed other advanced economies in terms of trade since 2016. We estimate that goods trade volumes would be 10-15% higher in a counterfactual in which volumes had grown in line with the average of other G7 economies, reflecting higher trade frictions outside of the EU (Exhibit 4, left). Services trade has held up better. Recent studies, however, also point to a significant hit to services trade as a result of leaving the EU (Exhibit 4, right).

Exhibit 4: UK Trade Has Been Hit

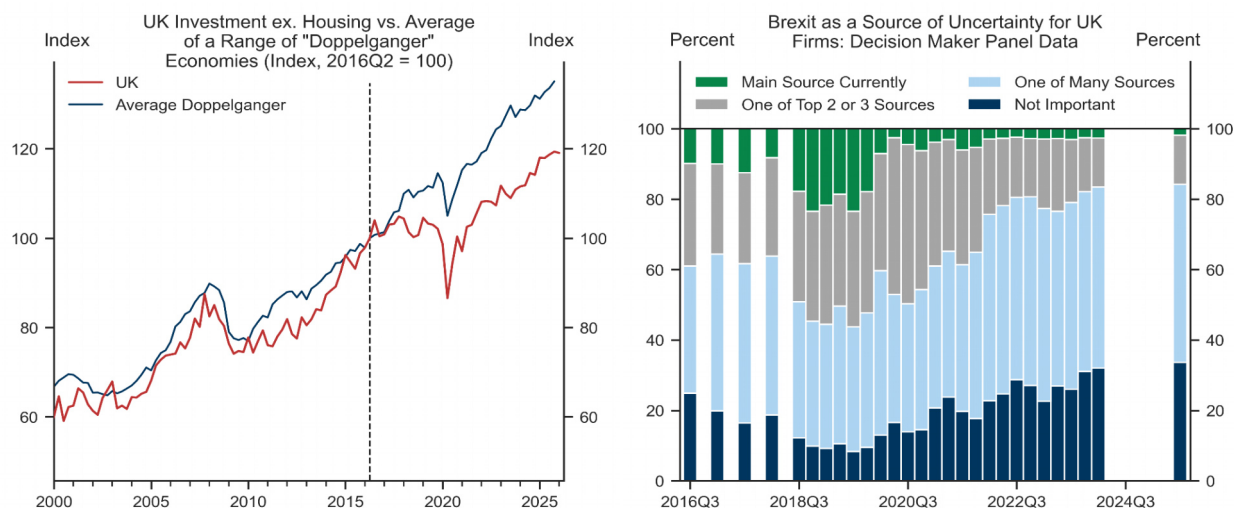
Source: Goldman Sachs Global Investment Research, Haver Analytics, Bhalotia et al. (2025)

Overall, the trade estimates point to a roughly 5-10% decline in trade volumes due to Brexit. Previous studies suggest that a decline in trade intensity of this magnitude could have reduced GDP by 2-4% through reduced productivity.²

Second, UK investment growth stalled after the referendum, falling significantly below its pre-2016 trend (Exhibit 5, left). Our Doppelgänger estimates for investment point to a shortfall of around 10% relative to other economies. At least some of this underperformance likely reflected Brexit uncertainty after the referendum, with surveys clearly flagging Brexit as a source of uncertainty (Exhibit 5, right). Our estimates on the business investment channel imply an output hit of just over 2% by end-2025.³

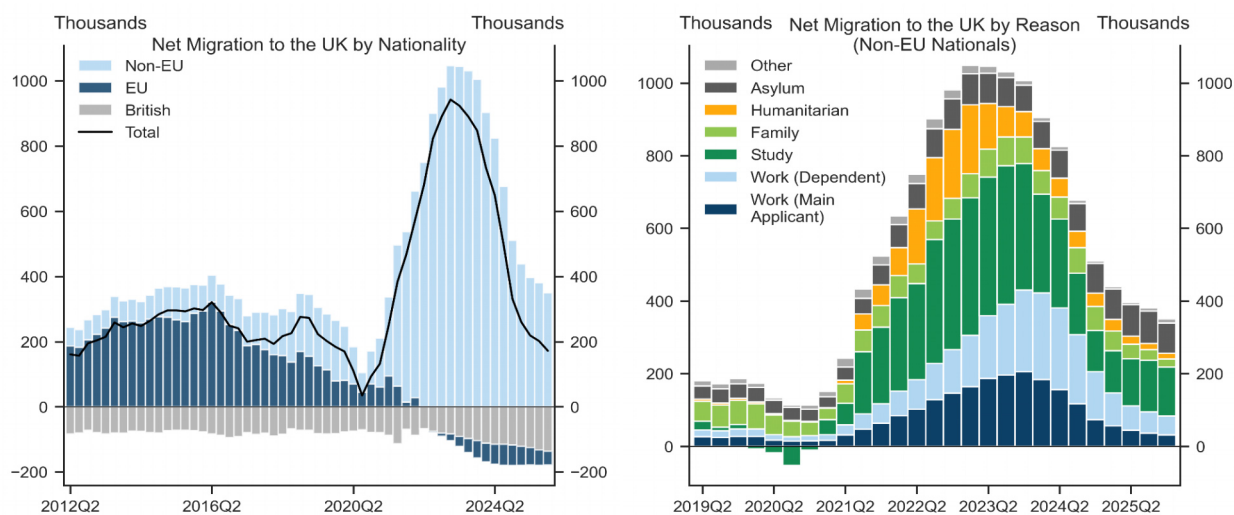
² Two studies by Feyrer using different methodologies have estimated that each 1% reduction in trade lowers GDP by around $\frac{1}{4}\%$ or $\frac{1}{2}\%$.

³ Haskel and Martin (2023) estimated that weakness in business investment after the referendum had lowered output by 1.3% by 2022.

Exhibit 5: Investment Has Underperformed

Source: Goldman Sachs Global Investment Research, Haver Analytics, Bank of England

Third, Brexit has affected immigration flows into the UK. Net migration from the EU has fallen sharply from a peak of over 300k in the year to 2016Q2 to -42k in 2025 (Exhibit 6, left). At the same time, net migration from outside the EU rose significantly to over 1m per year after the pandemic. Although much of the pickup in non-EU immigration reflects reasons other than work (Exhibit 6, right), recent [research](#) suggests that the new immigration system has so far led to a modest net increase in the number of foreign-born people in work. That said, net migration has dropped sharply recently, partly reflecting new government policies.

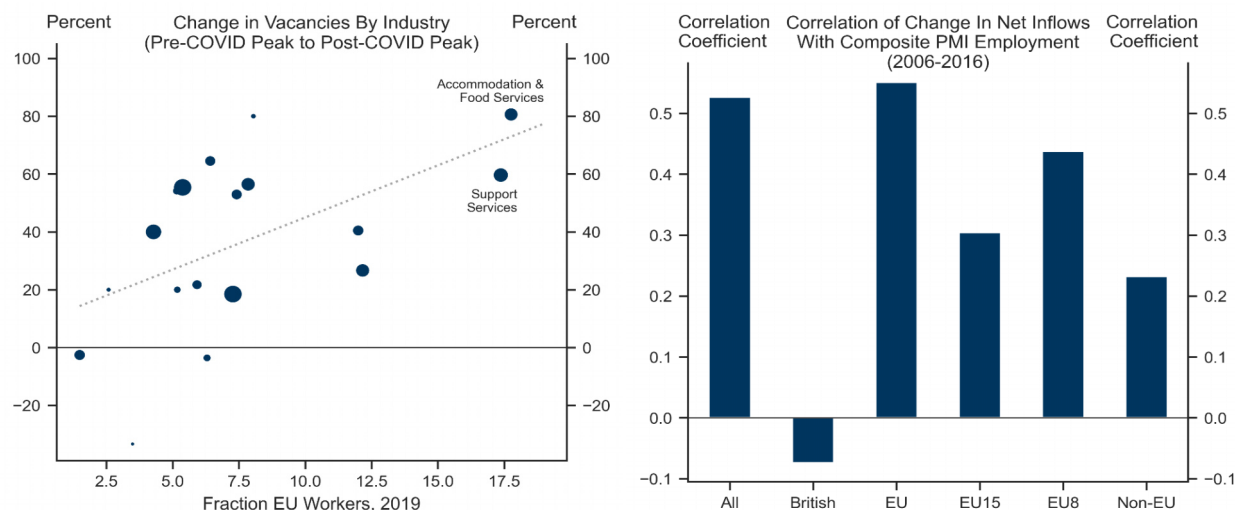
Exhibit 6: Net Migration from the EU Has Fallen but Overall Inflows Rose Post-Pandemic

Source: ONS

Despite the rise in net migration after Brexit, our previous [work](#) suggests that changes in the immigration system have probably contributed to labour shortages. EU nationals have historically been concentrated in lower-skilled occupations and are harder to replace under the new skill-selective migration regime. The sectors that are most reliant on EU workers have experienced the largest increases in vacancy rates since the Covid

pandemic (Exhibit 7, left). Going forward, reduced cyclical volatility of migration flows compared with the pre-Brexit period (Exhibit 7, right) could lead to greater volatility in labour market tightness and domestic inflationary pressures.

Exhibit 7: Lower Immigration from the EU Helps Explain Labour Shortages



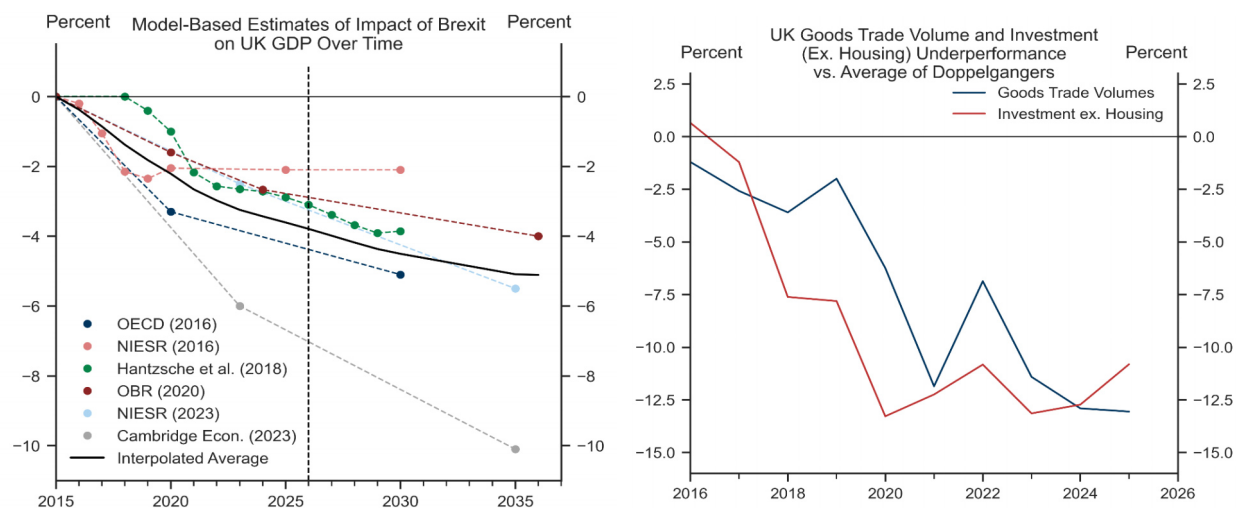
Source: Goldman Sachs Global Investment Research, Haver Analytics, ONS

As Bad as It Gets?

Taken together, our estimates show that Brexit has materially weighed on Britain's economic performance relative to other advanced economies. While new trade agreements could, in principle, help mitigate some of the costs of Brexit, the benefit from new (non-EU) trade deals is likely to be small.

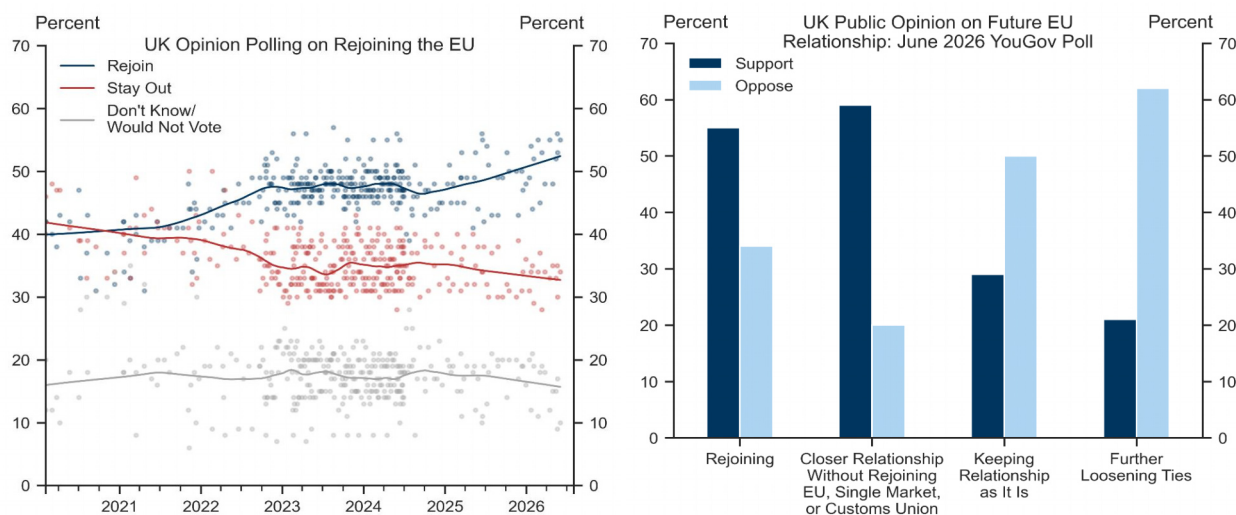
That said, we see two reasons to think that the worst of the Brexit headwind might now be behind us.

First, although the full impact of leaving the EU is likely to take time to materialise, several studies point to a front-loaded growth drag (Exhibit 8, left). Decision Maker Panel data and our own analysis indicate that much of the impact on investment had already come through before the UK left the EU at the end of 2020 as businesses anticipated future disruption to trade barriers. Although there is evidence that trade flows can take 7-10 years to adjust fully to new trade barriers, the shortfall in goods trade volumes relative to other economies shows some signs of levelling off in the last two years (Exhibit 8, right). It is true that the UK's underperformance on GDP has continued to build, perhaps reflecting lags from lower investment and trade into productivity. But these factors suggest that the growth hit is likely to gradually ease over time.

Exhibit 8: Past the Peak Brexit Growth Headwind

Source: Goldman Sachs Global Investment Research, Haver Analytics, Various academic studies

Second, recent opinion polls and political shifts point to building support for a closer EU-UK relationship. Public opinion polls show that a majority of the UK population now support rejoining the EU (52%), up notably since Covid and now comfortably ahead of the 33% share supporting Brexit (Exhibit 9, left). Public support is highest for a closer relationship with Europe without rejoining the EU, Single Market or Customs Union (Exhibit 9, right). As a result, the government has worked towards a “reset” in EU-UK relations and Andy Burnham (who is likely to succeed Prime Minister Starmer) has historically supported closer integration with Europe.

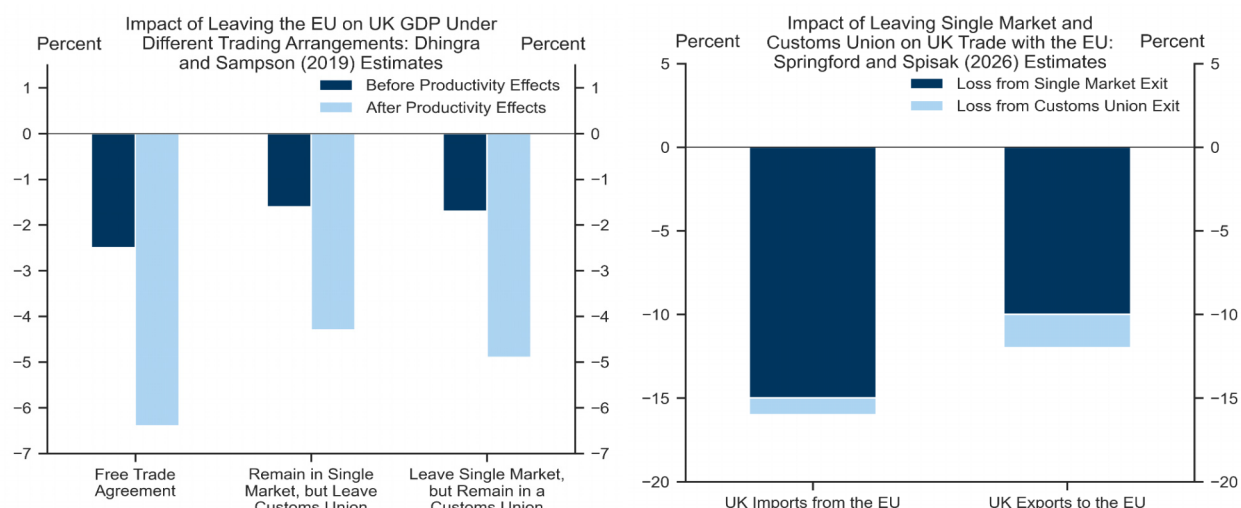
Exhibit 9: Public Opinion Has Shifted in Favour of a Closer Relationship

Source: Goldman Sachs Global Investment Research, Various polling companies, YouGov

The government could start by finalising the veterinary agreement, negotiating mutual recognition of professional qualifications, and aligning with EU rules where it is of clear economic benefit (e.g., for chemicals or the automotive sector). A more ambitious reset further down the line—such as rejoining the single market or negotiating a customs union—could raise GDP by 1-2% and thereby undo a meaningful share of the economic

cost incurred by Brexit (Exhibit 10).

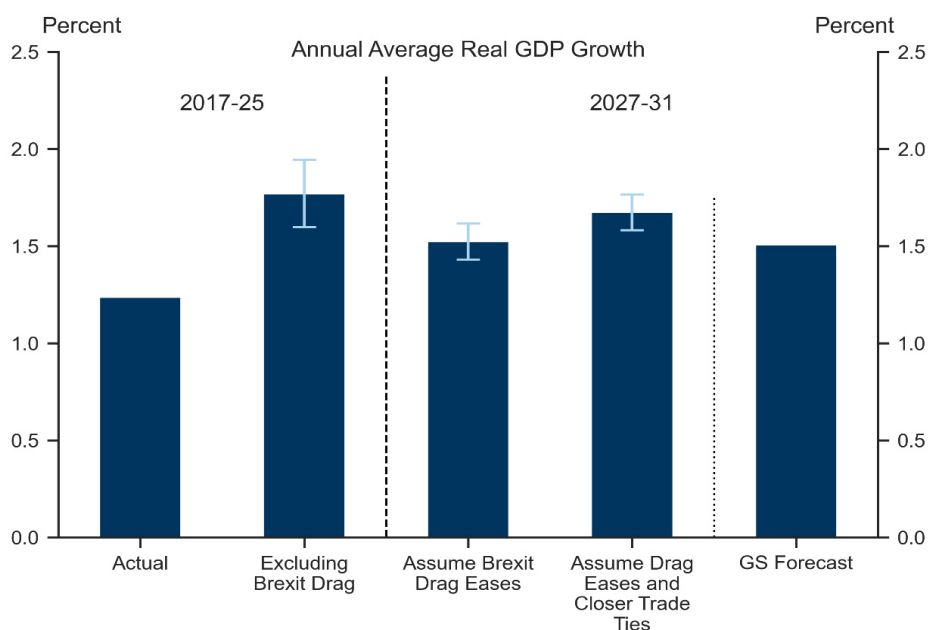
Exhibit 10: A Closer Economic Relationship Could Mitigate Some of the Brexit Costs



Source: Dhingra and Sampson (2019), Springford and Spisak (2026)

The combination of a receding Brexit headwind and a closer economic relationship with the EU could improve the UK's growth performance significantly in coming years (Exhibit 11). Simple calculations suggest that—assuming an unchanged underlying growth pace—UK trend growth could pick up to 1.5% in coming years with a diminishing Brexit drag. That figure could rise to 1.7% in the event of a significant reset of EU-UK relations.

Exhibit 11: An Easing Brexit Drag Could See Trend Growth Improve



“Excluding Brexit drag” uses the average of the GS doppelganger estimate and the growth headwind implied by the model-based estimates shown in Exhibit 8 (left). The light blue bracket shows the range from those two approaches. “Assuming Brexit Drag Eases” assumes that the drag fades in line with the time profile from the model-based estimates. “Closer Trade Ties” assumes a boost of 1-2% from a more ambitious reset over 10 years.

Source: Goldman Sachs Global Investment Research, Haver Analytics

While there is considerable uncertainty around the exact timing and magnitude of that improvement, the likely scale of the drag on growth over the past decade and the potential upside from a closer EU-UK relationship suggests that the upside is material. Consistent with this, we expect average growth in the UK to pick up from 1.2% since 2016 to 1.5% over the coming years.

James Moberly

Sven Jari Stehn

Appendix

Exhibit 12: Studies Estimating the Impact of Brexit on UK GDP

Model-Based Estimate of Long-Run Impact of Brexit			Tracking Estimates of the Impact of Brexit on GDP to Date		
HM Treasury	2016	6.2% under a bilateral agreement over 15 years	Born et al.	2019	1.7-2.5% by the end of 2018
OECD	2016	3.3% by 2020, 5.1% by 2030	Springford (CER)	2019	2.5% by the end of 2018
NIESR	2016	2.1% under FTA ex. productivity; productivity effects could increase cost significantly further	Fetzer and Wang	2020	2.3-2.5% aggregate impact by 2019, 8.5% in worst affected regions
Dhingra et al. (LSE/CEP)	2017	1.3-2.7% ex. productivity; 6.3-9.4% inc. productivity	Springford (CER)	2022	5.5% by 2022Q2
IMF	2018	2.6-3.9% under FTA, average of 3.1%	Alabrese et al.	2026	7.7% by 2023
Hantzsche et al. (NIESR)	2018	3.9% under May deal	Bloom et al.	2026	6-8% by end of 2025
Dhingra and Sampson (LSE/CEP)	2019	2.5% ex. productivity, 6.4% inc. productivity under Johnson deal	NIESR	2026	4.9-9.0% by end 2025, depending on comparison group of countries
Hantzsche and Young (NIESR)	2019	3.5% under Johnson deal			
OBR	2020	4% in the long-run under an FTA			
NIESR	2023	2-3% by 2023, 5-6% by 2035			
Cambridge Econometrics	2023	6.0% by 2023, 10.1% by 2035			

Source: Data compiled by Goldman Sachs Global Investment Research, Various academic studies

European Economics Team

Sven Jari Stehn

+44(20)7774-8061
jari.stehn@gs.com
Goldman Sachs International

James Moberly

+44(20)7774-9444
james.r.moberly@gs.com
Goldman Sachs International

Giovanni Pierdomenico

+44(20)7051-6807
giovanni.pierdomenico@gs.com
Goldman Sachs International

Filippo Taddei

+44(20)7774-5458
filippo.taddei@gs.com
Goldman Sachs International

Niklas Garnadt

+49(69)7532-1537
niklas.garnadt@gs.com
Goldman Sachs Bank Europe SE

Alexandre Stott

+33(1)4212-1108
alexandre.stott@gs.com
Goldman Sachs Bank Europe SE - Paris Branch

Katya Vashkinskaya

+44(20)7774-4833
katya.vashkinskaya@gs.com
Goldman Sachs International

Disclosure Appendix

Reg AC

We, James Moberly and Sven Jari Stehn, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: James Moberly Goldman Sachs International, Sven Jari Stehn Goldman Sachs International.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered

companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g.,

marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

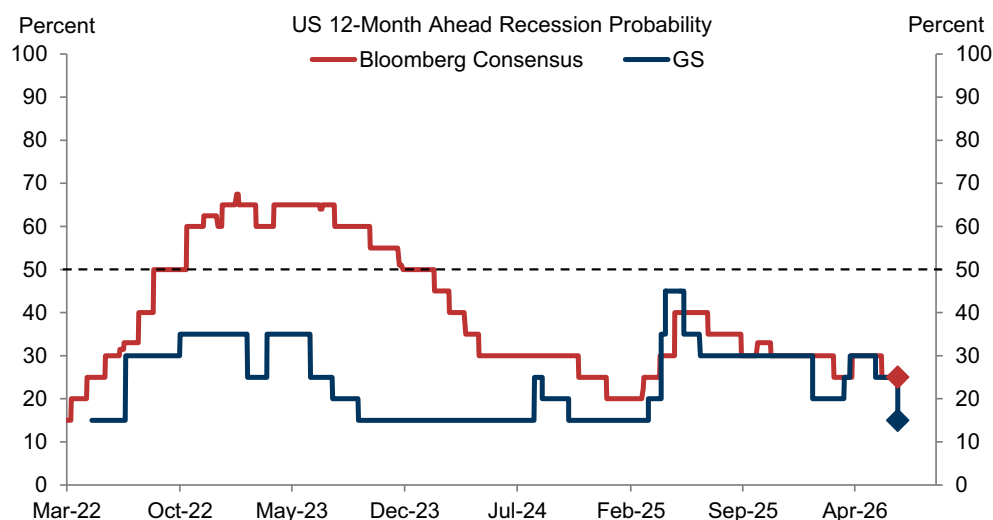
You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.

Global Views: More Crude, Less Concern

1. The agreement between the US and Iran has reduced the downside risks to our economic outlook. Our commodities strategists now see Brent crude at \$80 per barrel by the end of 2026, with two-sided risks. On the upside, Iran's announcement on Saturday that the Strait was closed again served as a reminder that oil flows might only recover slowly. On the downside, a near-term glut could develop as oil is released quickly into a market that was already oversupplied before the war. The agreement has led us to cut our 12-month US recession risk estimate further from 25% to the long-term norm of 15%. (This is below our 20% estimate on the eve of the war because the labor market improvement since then indicates greater underlying resilience.)

Jan Hatzius
+1(212)902-0394 | jan.hatzius@gs.com
Goldman Sachs & Co. LLC

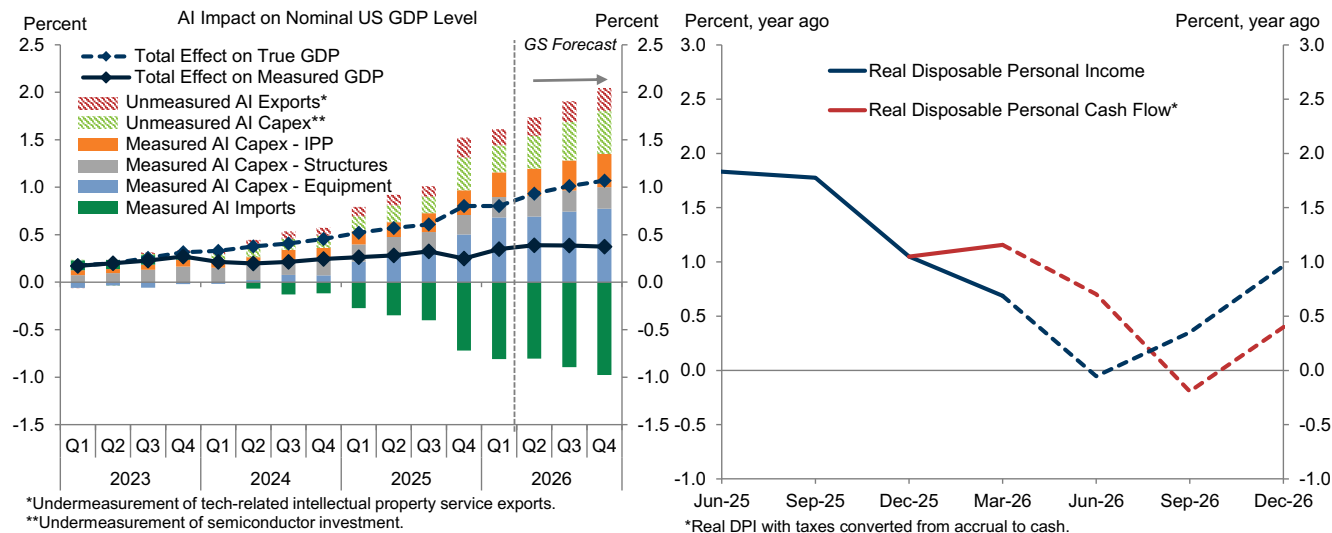
Exhibit 1: US Recession Risk Back to the Long-Term Norm of 15%



Source: Bloomberg, Goldman Sachs Global Investment Research

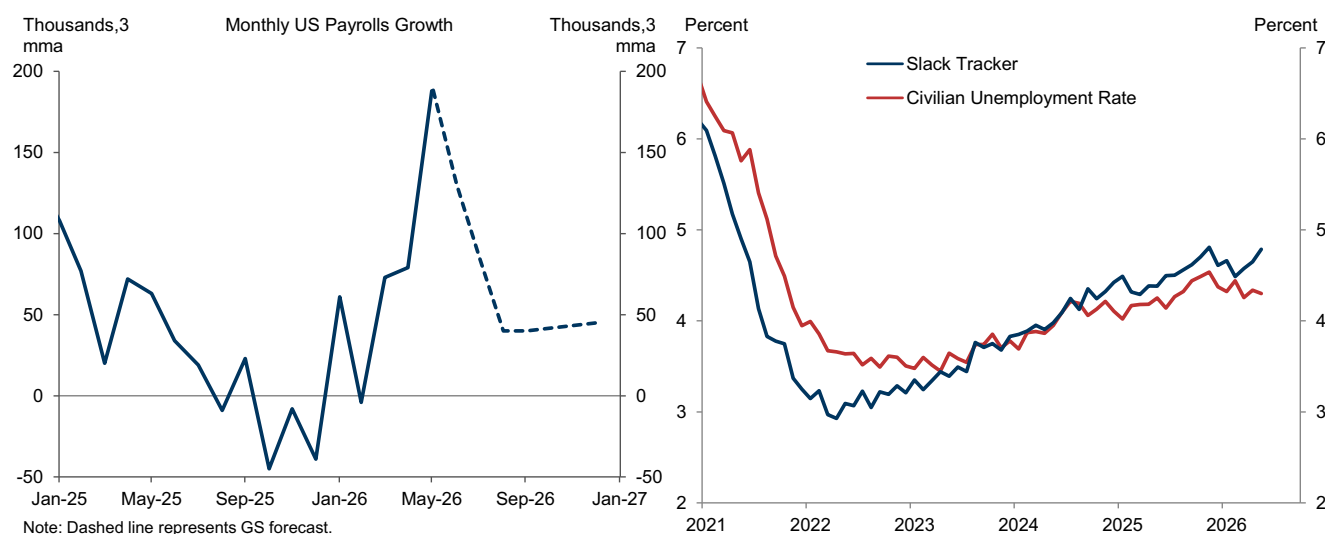
2. Alongside the cut to our recession risk estimate, we have nudged up our sequential H2 GDP growth forecast to 2%. The slightly stronger path reflects a positive sequential impulse to real income from lower gas prices, at a time when the economy continues to benefit from the AI boom via higher equity wealth as well as strong capex. But we still expect growth to remain moderate. First, AI capex largely consists of goods that are imported from Asia and, in the case of semiconductors, don't even show up in GDP. Second, even with the drop in gas prices, real income defined on a cash flow basis is likely to slow in H2 as the impact of higher tax refunds and lower final tax payments in Q2 peters out. This is likely to keep real consumer spending growth at just 1½%.

Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html.

Exhibit 2: AI Investment Surge Adds Little to US GDP and the US Consumer Remains Under Pressure

Source: Bureau of Economic Analysis, Haver Analytics, Goldman Sachs Global Investment Research

3. With GDP still growing modestly below potential, we expect a slowdown in payroll gains from the outsized 188k rate over the past three months to just below our 60k breakeven estimate. Beyond the growth forecast, we would note the failure of other labor market indicators to confirm a sharp recent improvement. Household employment has grown much more slowly than payrolls, the employment/population ratio has trended sideways to lower, and our updated slack tracker—which summarizes ten key indicators of labor market utilization—has recently resumed its gradual loosening of the prior three years. Consistent with this, we expect wage and unit labor cost growth to remain muted despite the earlier increase in headline inflation.

Exhibit 3: The Payroll Surge Is an Outlier Relative to Other Labor Market Indicators

Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs Global Investment Research

4. The plunge in gasoline prices so far in June should result in an outright decline in seasonally adjusted consumer prices. We also expect core CPI inflation to average just 0.17% in the next three months as airfares decline on the back of lower jet fuel prices, hotel rates—which are measured at the time of booking—drop from their World Cup

highs, and rent inflation reverses what looks like a high-side outlier in the Northeast in May. Core PCE inflation is likely to remain more stubborn, in part because of ongoing upward pressure on imputed financial services prices from the equity market rally and price increases in software/accessories whose weight in core PCE is 30 times larger than in core CPI. More statistically based measures such as trimmed-mean PCE should continue to look more benign, however.

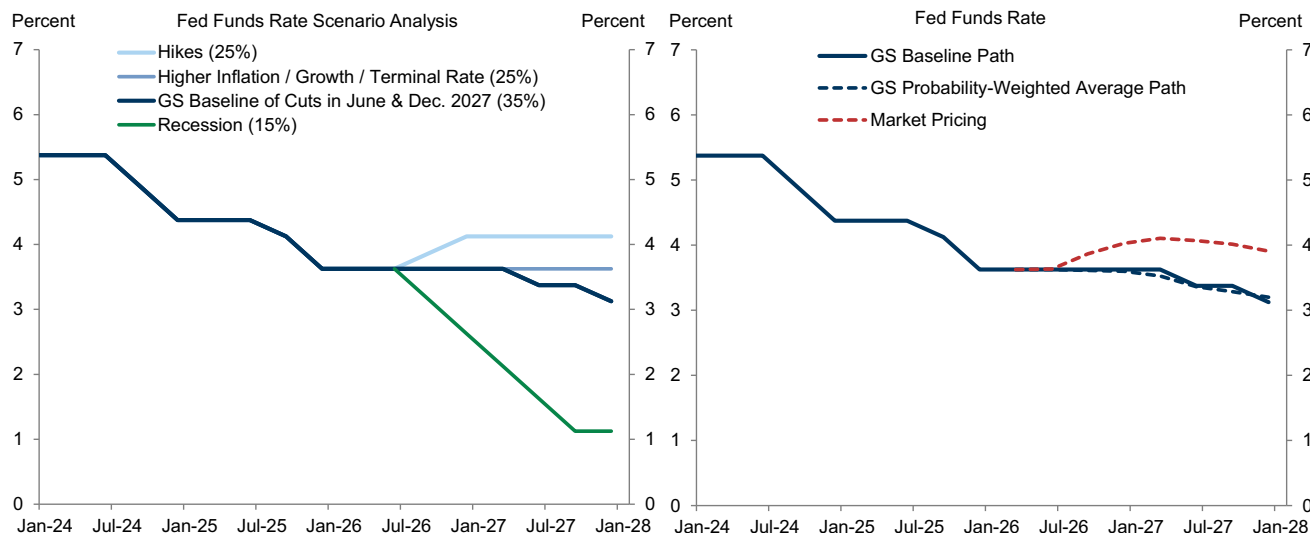
Exhibit 4: Trimmed-Mean PCE Still Looks More Benign Than Core PCE



Note: May 2026 values are GS forecasts based on inflation tracking estimates.

Source: Haver Analytics, Goldman Sachs Global Investment Research

5. The first FOMC meeting under Chairman Warsh was more hawkish than expected. Half of the 18 participants that submitted interest rate projections penciled in one or more rate hikes in the remainder of 2026, and the median projection for core PCE inflation in 2027Q4 rose to 2.5%. While this has increased the upside risks, our baseline forecast remains for no hikes. First, we suspect that about half of the 2026 hikers are nonvoting Federal Reserve Bank presidents, which would leave a majority of voters projecting unchanged or lower rates under appropriate policy. Second, the projections might be stale given the rapid improvement in the news from the Middle East and the plunge in energy prices over the past week. If inflation comes down in coming months—rapidly for headline and gradually for core—and growth remains muted, we expect most voters to continue supporting unchanged rates.

Exhibit 5: While Hikes Have Become More Likely, We Think They Are Less Likely Than Markets Are Pricing

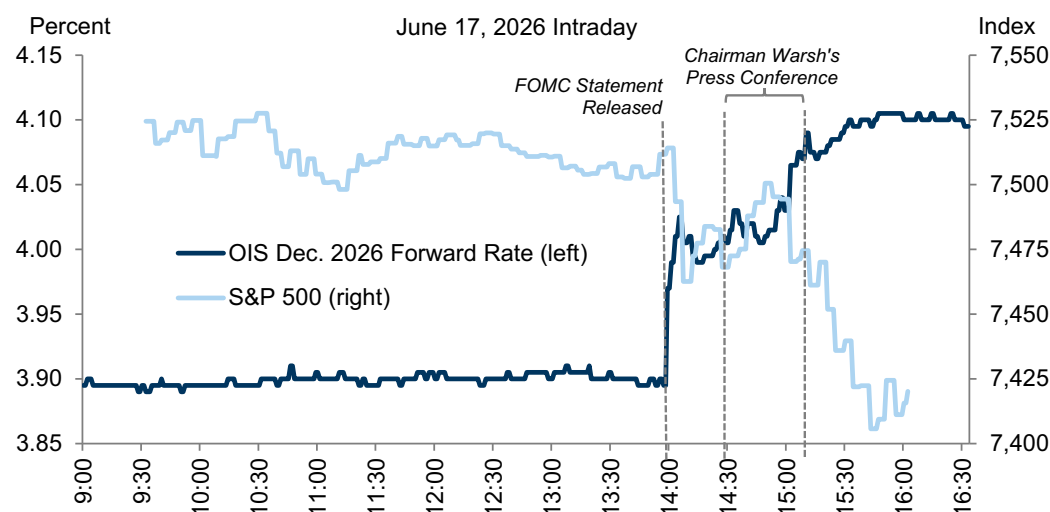
Source: Bloomberg, Goldman Sachs Global Investment Research

6. We had expected Warsh's refusal to provide either a "dot" or verbal guidance about upcoming rate decisions but his broader comments on Fed communications with financial markets went further:

"So I think financial markets perform best when they react to incoming data. I think the financial markets work less efficiently when they ask [the] question 'how will the Federal Reserve react to that incoming information?' The more that markets are paying attention to what's happening in the real economy, deciding what's good data and what's less good data, the more financial markets can price what they believe is the most likely and what are the tail risks."

Since the Fed essentially controls short-term interest rates and the interest rate path matters for asset pricing, it is inevitable that markets ask how the Fed will react to incoming information. Admittedly, this can turn into an unproductive obsession with computing the Fed's reaction function, which is often harder to summarize and subject to greater change than many market participants think. Still, a significant reduction in Fed transparency could make financial conditions—and thus economic outcomes—more volatile as market assumptions about the Fed's thinking become subject to fewer reality checks.

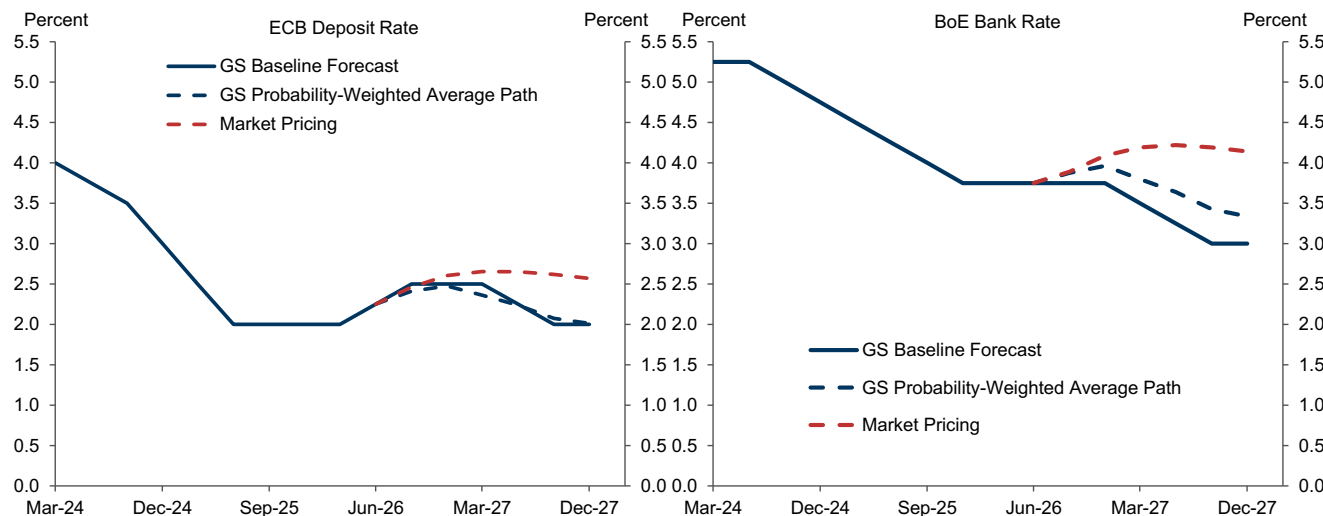
Exhibit 6: FOMC Meeting Illustrates the Importance of Fed Communications for Financial Conditions



Source: Bloomberg, Goldman Sachs Global Investment Research

7. The ECB hike to 2.25% was widely expected, but the drop in energy prices has shifted the balance of risks around our forecast of one more 25bp move in September to the downside. Either way, we continue to expect a return to a 2% deposit rate in 2027 and thus remain dovish relative to market pricing. In the UK, Sir Keir Starmer has stepped down as prime minister in favor of Andy Burnham, who inherits an economy with more potential than widely appreciated. On monetary policy, the improvement in wage and price inflation even prior to the recent oil price drop has strengthened our view that the MPC won't hike in coming months and will ultimately cut Bank Rate to 3% in 2027, well below market pricing.

Exhibit 7: Our ECB and BoE Rate Views Remain Below Market Pricing

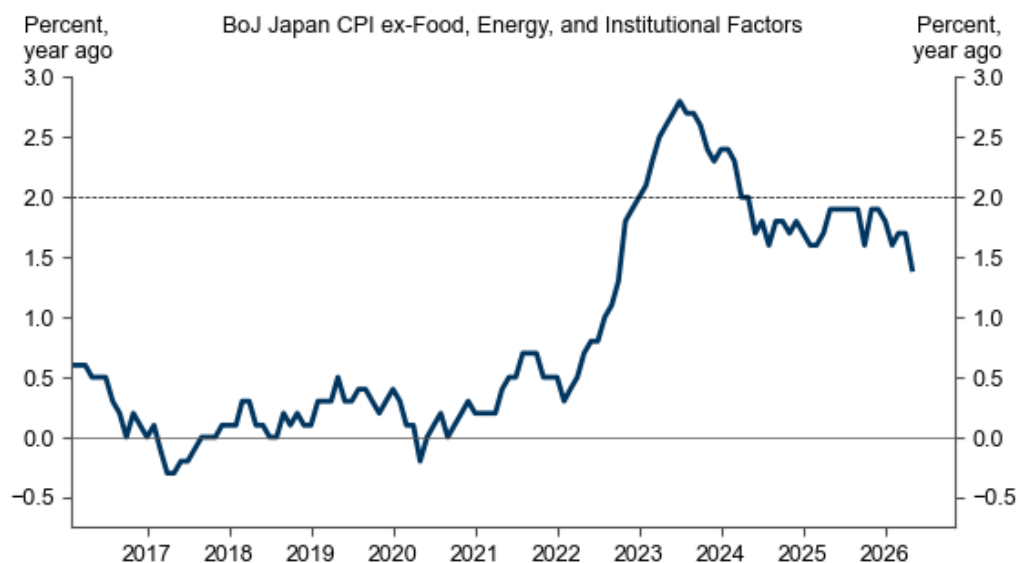


Source: Bloomberg, Goldman Sachs Global Investment Research

8. The Bank of Japan hiked by 25bp to 1%, the highest level in 30 years. We expect two more 25bp hikes every six months or so until the policy rate hits 1½% next summer. The BoJ can take its time because inflation—using its estimate of CPI excluding all food and

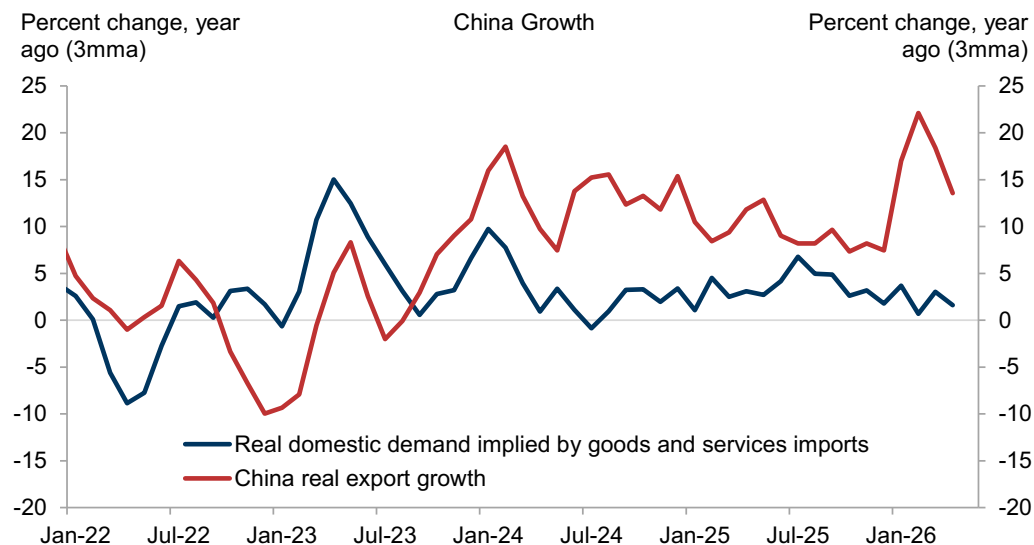
energy as well as institutional factors such as changes in school tuition rates—is running at only about 1½% year-on-year and GDP is likely to grow just 0.5% in 2026. The biggest challenge for the Japanese economy remains fiscal sustainability, as the decline in the debt/GDP ratio over the past decade could reverse if Prime Minister Takaichi implements her expansionary plans and long-term interest rates climb further toward the levels seen in other G10 economies.

Exhibit 8: Underlying Inflation in Japan Remains Benign



Source: BoJ, Haver Analytics, Goldman Sachs Global Investment Research

9. We attribute most of the downside growth surprises in China in April and May to a combination of the oil shock, payback for the consumer goods trade-in program, and the government's decision to slow fiscal spending after the strong Q1 GDP print. June is likely to remain sluggish because of heavy rainfall, but we expect a rebound in Q3 as the temporary drags abate, the oil shock reverses, and policymakers respond to the recent weakness with more fiscal spending. Underneath the short-term ups and downs, the gap between domestic demand and exports continues to grow, which drives up the current account surplus with negative effects on many of China's trading partners.

Exhibit 9: China's Domestic Demand Is Growing Far More Slowly Than Its Exports

Source: Haver Analytics, Goldman Sachs Global Investment Research

10. US financial markets have digested the hawkish FOMC shock well, with declines in breakeven inflation and 30-year Treasury yields alongside a rebound in broad equity indices. If the near-term economic outlook settles into calmer waters, the focus on the sustainability of the AI trade is likely to grow even further. The work of our equity and cross-asset strategists suggests that ongoing upside surprises in AI capex, strength in S&P earnings, and the relative absence of private sector financial imbalances should all help equity markets make further headway in coming months. However, valuations of AI-related stocks are now high even relative to our optimistic views regarding the economic value that is likely to be created by AI over the next decade, so it is getting harder to justify further large gains.

Jan Hatzius

Disclosure Appendix

Reg AC

I, Jan Hatzius, hereby certify that all of the views expressed in this report accurately reflect my personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Jan Hatzius Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered

companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g.,

marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.

US Daily: June FOMC Recap: Hikes Look More Likely but Are Still Not Our Base Case

- The June FOMC meeting delivered a hawkish surprise, with nine participants projecting a hike in 2026 (vs. our expectation of three) and the median projection for 2027 Q4/Q4 core PCE inflation rising to 2.5% (vs. our expectation of 2.3%). Market pricing for the fed funds rate at the end of 2026 rose by 20bp in response to the projections and Chairman Kevin Warsh's press conference.
- Today's meeting raises the risk of interest rate hikes later this year, but our base case is still that the FOMC will leave the policy rate unchanged this year. A majority could support a hike, perhaps for a range of different reasons, if upcoming inflation prints are uncomfortable and job growth remains solid. But we suspect that most of the voters still lean toward an unchanged policy rate and that participants treated the news about a deal with Iran and the reopening of the Strait of Hormuz cautiously for now. If so, today's projections could prove stale if trade resumes, which should resolve the most serious source of upside risk to inflation fairly quickly.
- Warsh also announced that he is creating five task forces to reconsider Fed communications, the Fed's balance sheet, the Fed's use of and reliance on existing data sources, productivity and jobs in "an era of transformation," and the Fed's inflation frameworks. He appeared particularly focused on rethinking Fed communications. Fed officials have already discussed some of these topics recently, and it is hard to know if the task forces will reach different conclusions.

David Mericle
+1(212)357-2619 |
david.mericle@gs.com
Goldman Sachs & Co. LLC

June FOMC Recap: Hikes Look More Likely but Are Still Not Our Base Case

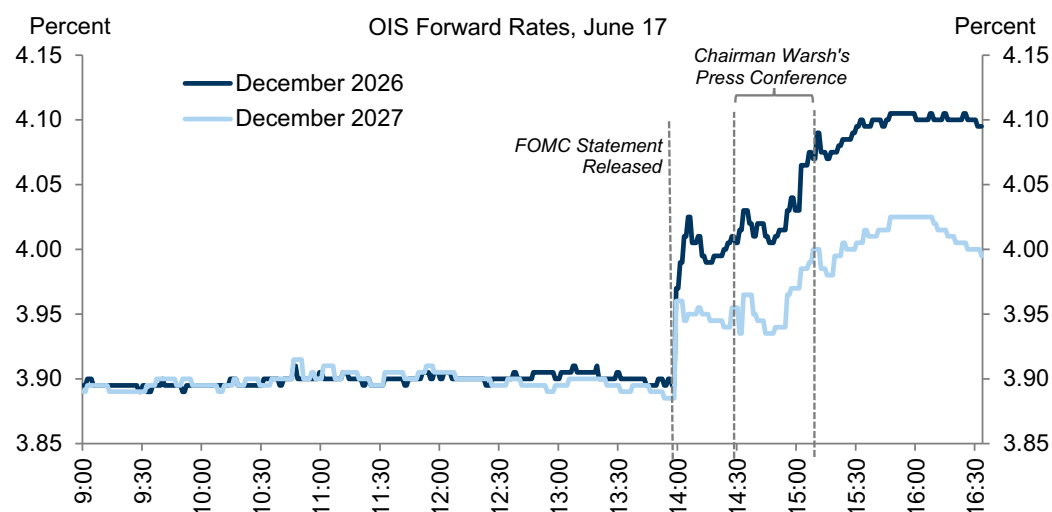
As widely expected, the FOMC left the funds rate unchanged at 3.5-3.75% today and removed the previous forward guidance suggesting cuts from its statement at the June FOMC meeting today. The FOMC also substantially shortened and simplified its post-meeting statement.

But the meeting also delivered a significant hawkish surprise, with nine participants projecting a rate hike in 2026, against our expectation that just three would. Because the new Chairman Kevin Warsh chose not to submit projections, this resulted in a nine to nine tie in the dot plot between those projecting at least one hike and those projecting an unchanged policy rate or, in one case, a rate cut.

Consistent with the hawkish message from the dots, the key surprise in the economic projections was that the median forecast for 2027 Q4/Q4 core PCE inflation rose 0.3pp to 2.5%, above our expectation that it would rise more modestly to 2.3%. This suggests that participants were more concerned that the recent firmer inflation news, caused in part by the energy cost shock from the war with Iran, has implications for inflation further down the road.

It was not clear where Warsh stood on the likelihood of a hike being appropriate from the press conference. He took a balanced view on whether the current stance of monetary policy is restrictive, noting that it is likely restraining housing activity but it is hard to say the same of the impact on financial markets. But he repeatedly emphasized the Fed's commitment to price stability, which might make him sympathetic to a hike if upcoming inflation reports are discouraging. And he said that "financial market prices are probably the most important source of information to guide central bankers," which might imply that he sees market pricing of a higher funds rate later this year as supporting the case for hiking.

In response to both the projections and Warsh's press conference, market pricing for the fed funds rate at the end of 2026 rose by 20bp and pricing for the end of 2027 rose by 11bp (Exhibit 1).

Exhibit 1: The Bond Market Took Both the FOMC's Projections at 2pm and Chairman Warsh's Press Conference Starting at 2:30 as Substantial Hawkish Surprises


Source: Goldman Sachs Global Investment Research

Today's meeting raises the risk of interest rate hikes later this year, but our base case is still that the FOMC will leave the policy rate unchanged this year. A majority could support a hike, perhaps for a range of different reasons, if upcoming inflation prints are uncomfortable and job growth remains solid. Some might be more concerned about recent inflation trends, while others might be less convinced that a hike is necessary but see little harm from a starting point where job growth has been running strong. A great deal will depend on upcoming inflation reports, and we do expect that May core PCE inflation will be firm at 31bp, though 12bp of that would come from financial services.

But we suspect that a majority of the twelve voters, as opposed to the nineteen participants, still lean toward an unchanged policy rate. We also suspect that participants treated the news about a deal with Iran and the reopening of the Strait of Hormuz cautiously for now. If so, today's projections could prove stale if trade resumes, which should resolve the most serious source of upside risk to inflation fairly quickly. Warsh noted that "my colleagues around the table, when they submitted their dots, understand the world is changing quite quickly" and will not feel bound by their projections.

Warsh also announced that he is creating five task forces to investigate Fed communications, the Fed's balance sheet, the Fed's use of and reliance on existing data sources, productivity and jobs in "an era of transformation," and the Fed's inflation frameworks. The task forces will include Fed staff economists as well as people from outside the Fed, including non-economists.

Warsh appeared particularly focused on reconsidering Fed communications and especially on making changes to the Summary of Economic Projections. He also said that "press conferences are useful, but when you have one you want to make sure you have something important to say," which could hint at a break from the current pattern of press conferences after each of the eight FOMC meetings.

Fed officials have already discussed some of the topics assigned to the task forces

recently. As Warsh noted, the FOMC discussed its communications practices as part of its 2025 framework review. It also discussed balance sheet policy as part of its 2020 framework review, and the Fed staff have written a number of research notes on the balance sheet recently. At this early stage, it is hard to know if the new task forces will reach different conclusions and ultimately produce meaningful policy changes.

David Mericle

The US Economics Team

Jan Hatzius

+1(212)902-0394
jan.hatzius@gs.com
Goldman Sachs & Co. LLC

Alec Phillips

+1(202)637-3746
alec.phillips@gs.com
Goldman Sachs & Co. LLC

David Mericle

+1(212)357-2619
david.mericle@gs.com
Goldman Sachs & Co. LLC

Ronnie Walker

+1(917)343-4543
ronnie.walker@gs.com
Goldman Sachs & Co. LLC

Manuel Abecasis

+1(212)902-8357
manuel.abecasis@gs.com
Goldman Sachs & Co. LLC

Elsie Peng

+1(212)357-3137
elsie.peng@gs.com
Goldman Sachs & Co. LLC

Pierfrancesco Mei

+1(212)902-8809
pierfrancesco.mei@gs.com
Goldman Sachs & Co. LLC

Jessica Rindels

+1(972)368-1516
jessica.rindels@gs.com
Goldman Sachs & Co. LLC

Disclosure Appendix

Reg AC

I, David Mericle, hereby certify that all of the views expressed in this report accurately reflect my personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: David Mericle Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered

companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g.,

marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.